Lifting Businesses™

CHE simulator user manual

1 INTRODUCTION

This manual instructs how to operate with che simulator application. Che simulator offers simple interface to TOS system including these features:

- connecting to TOS system as client or server
- sending pick/place messages to TOS system
- sending heartbeat messages to TOS messages
- receiving job messages from TOS system
- actions window to follow message flow

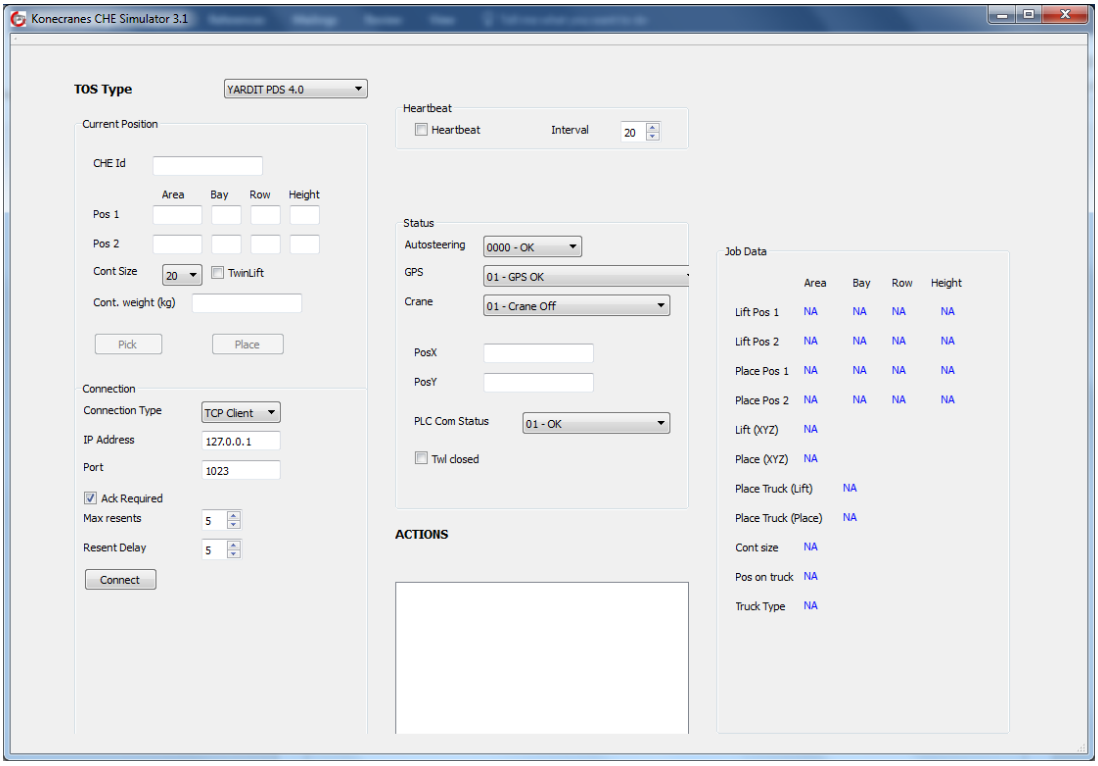
Before using che simulator it has to be installed by executing corresponding installer. It is a window executable (KcCheSimulatorInstaller.exe) and it creates startup icon to desktop.

Icon that installer creates looks like this.

Start your application double clicking this icon.

2 User interface

User interface window looks like this



From **TOS Type** dropdown box you can select between three different versions

YARDIT PDS 4.0
YARDIT PDS 3.3
YARDIT PDS 3.0

Different sections of the user interface are handled in the next chapters.

2.1 Connection section

In **Connection Type** dropdown box you can select between TCP Client and TCP Server options. Other options are currently not working.

IP Address is the destination (TOS) computer IP address. If TCP Server is used, then address selection is not meaningful and is neglected because TCP server is listening at any port.

Port is the port where TOS system is listening (Client Mode) or where CHE simulator is listening (Server Mode).

If **Ack Required** check box is checked then job message is confirmed with acknowledge message. Also pick/place message expects acknowledgement.

Max Resents spin box used to set up how many times pick/place message is resent if not acknowledged. This setting is meaningful only if **Ack Required** check box is checked.

Resent Delay spin box is used to set up how many times pick/place message is resent if not acknowledged. This setting is meaningful only if **Ack Required** check box is checked.

Connect/Listen push button is used to connect to tcp server (client mode) or to start listening at predefined port (server mode).

2.2 Current position section

At **CheId** text box the name of the simulated crane can be given.

At **Pos1** the position of the container 1 can be given.

At **Pos2** the position of container 2 (only twinlift) can be given.

Container size can be selected from dropdown list **Cont size.**

Twinlift can be activated by checking corresponding checkbox.

Cont. Weight (kg) text box can be used to feed in container weight in kilograms.

By pressing **Pick** or **Place** push buttons pick/place messages can be sent.

2.3 Heartbeat section

Heartbeat checkbox can be used to activate heartbeat message**.**

Heartbeat message will be sent at constant intervals defined at **Interval** selection box. Number defines the interval in seconds.

2.4 Status section

In this section the following data can be given to be sent in heartbeat message:

- Autosteering Status (OK/ERROR)
- GPS Status (selection between seven different statuses)
- Crane Status (selection between four different statuses)
- Crane X position in meters compared to base station
- Crane Y position in meters compared to base station.

- PLC communication status (OK/Failure)

2.5 Job data section

In this section job data message information is displayed including

- Lift position 1 for container 1
- Lift position 2 for container 2 (only twinlift)
- Pick position 1 for container 1
- Pick position 2 for container 2 (only twinlift)
- Lift xyz position (if used)
- Pick xyz position (if used)
- Truck place for lift (if used)
- Truck place for pick (if used)
- Container size in feet
- Position on truck (if used)
- Truck Type (if used)

2.6 Actions window

In actions window messages flowing from/to simulator will be displayed most recent message always at top.